Data Quality Log

Project: Gross margin decline, Q1 → Q3 2026 · **Version:** v1.0 · **Author:** Ogunlusi Teniola Elizabeth, Data Analyst · **Date:** 17 October 2026
Data: six tables (`orders` , `order_lines` , `products` , `customers` , `supplier_costs` , `fx_rates`) · **Scope of the cleaned data:** 1,890 orders and 4,712 order lines, dated 1 Jan–28 Sep 2026

This log records every data defect found, how it was found, what it would have done to the numbers if left alone, how it was treated, and what is still open. Counts come from `01_profile.sql` and `05_validation_checks.sql` ; the supporting queries are in the appendix.

Severity: High = would push Q1 or Q3 outside Finance's tolerance (±0.5% revenue, ±0.2 points GM) · **Medium** = distorts months, groups or lines · **Low** = cosmetic or no effect on margin.

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1. Summary

ID	Issue	Amount	Severity	Treatment	Status
DQ-01	`order_lines` loaded twice	9,612 rows = 2 × 4,806	High	Extra copies removed by the de-duplication rule	Handled; reload recommended
DQ-02	Exact duplicate lines (sync bug)	94 lines; ₦248.6M net revenue	High	Keep lowest `line_id` ; remove the rest	Resolved
DQ-03	Mixed date formats	177 of 1,890 orders (9.4%) in dd/mm/yyyy	Medium	Parsed day-first	Resolved
DQ-04	Blank currency	107 lines	Medium	Treated as NGN	Resolved (Finance rule)

DQ-05	USD-priced lines among NGN lines	27 lines	High	Converted at the month's FX rate	Resolved
DQ-06	Rep-name spellings and blanks	6 spellings for 4 people; 88 orders blank	Low	Standardised; blank = "Unassigned"	Resolved
DQ-07	Bag-product quantities change unit	239 lines from 1 May	High	kg = qty × 25 from 1 May	Resolved; warehouse confirmation pending
DQ-08	Product-name formatting	e.g. `MALTODEXTRIN` (trailing space), `sorbitol 70%`	Low	Not changed; `product_id` is used	Open (cosmetic)
DQ-09	Lines that look alike but are not exact copies	714 lines; 26 distributor lines in Q3 unexplained	Medium	Kept, per Finance's rule	**Open** (IT confirmation)
DQ-10	Lines sold below cost (plausibility)	142 lines, all in Q3; ₦146.8M revenue	Medium	Kept; reported as a finding	Open (Finance to confirm intended)

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2. Detail

DQ-01 · `order_lines` loaded twice

- **Found by:** the first profiling run returned 9,612 rows for `order_lines`; both source files hold 4,806, with 4,806 distinct `line_id`s. Checks 1a, 1c and 1f in `01_profile.sql` read de-duplicated rows.
- **If untreated:** every line is counted twice. Revenue and cost both double, so **the margin % looks unchanged**: the error would not show in the margin itself.
- **Treatment:** `02_clean.sql` removes exact copies with `ROW_NUMBER()`. The same rule also removes the DQ-02 duplicates (4,806 load copies + 94 = 4,900 rows from 9,612; 4,712 remain).
- **Residual risk:** none for this analysis. Reloading `order_lines` once would remove the

need for the workaround on future refreshes.

DQ-02 · Exact duplicate lines (sync bug)

- **Found by:** check 1f (94). Finance: *IT confirmed a sync bug that double-posted some lines, and 94 sounds right.*
- **Rule (Finance):** exact copies only (same order, product, qty, price, discount, currency; only `line_id` differs); keep one; log the count and value.
- **Amount removed:**

Quarter	Lines	Net revenue ₦M
Q1	28	78.2
Q2	29	89.3
Q3	37	81.1
Total	**94**	**248.6**

- **If untreated:** Q1 net revenue ₦5,269.3M (**+1.5%** against Finance) and Q3 ₦3,906.7M (**+2.1%**), both outside tolerance.
- **Treatment:** the lowest `line_id` of each group is kept. The copies are identical, so this choice has no numeric effect.

DQ-03 · Mixed date formats

- **Found by:** check 1b: 1,713 orders in yyyy-mm-dd and **177 in dd/mm/yyyy**.
- **Proof of day-first:** 88 slash dates have a day above 12 and **none** has a month above 12.
- **If untreated (read month-first):** 88 dates fail to parse, 79 land silently in the wrong month, and only 10 (where day = month) come out right.
- **Treatment:** each format is parsed explicitly (`02_clean.sql`). Check 2e confirms every date is a real date; V08 confirms 0 invalid.

DQ-04 · Blank currency

- **Found by:** check 1c: NGN 4,672 · blank 107 · USD 27.
- **Basis:** Finance: *blank means NGN. The field was added later and was never mandatory.*
- **Treatment:** blank becomes NGN before any comparison or pricing, including inside the duplicate rule, so a blank-currency line and an identical NGN line count as copies.

DQ-05 · USD-priced lines

- **Found by:** check 1c (27 lines) and the price ranges: USD lines are priced 0.64–17.67; NGN lines 1,012.39–28,952.31.
- **Basis:** Finance: *lines tagged USD really are USD-priced; convert at that month's rate*.
- **If untreated (USD price read as naira), estimated:** Q1 net revenue ₦5,145.4M (**–0.9%**, GM 32.67%) and Q3 ₦3,805.5M (**–0.5%**), outside tolerance.
- **Treatment:** price × that month's FX rate (`03_margin_model.sql`).

DQ-06 · Rep names

- **Found by:** check 1d: `Chika O.` 322, `Funmi B.` 300, `IBRAHIM M.` 288, `Ibrahim M.` 302, `Tola A.` 292, `tola a.` 298, blank 88.
- **If untreated:** one rep appears as two in any rep-level view. No effect on margin totals.
- **Treatment:** standardised to four names; blank = "Unassigned" (88). The 88 blanks may be missing data and not typos; the label does not explain them.

DQ-07 · Bag-product quantities change unit

- **Found by:** reconciliation, not profiling. Q1 tied to Finance; Q3 was ₦3,565.3M against ₦3,825.6M (**–6.8%**). Finance had said bag products are recorded differently at some point and asked for proof from the data.
- **Products:** P03 Potassium Sorbate and P16 Sodium Bicarbonate (unit `bag_25kg`).
- **Evidence:** before 1 May every quantity is a multiple of 25 (25 to 3,000); from 2 May the quantities are 1, 2, 4, 8, 10, 20, 40, 80, 120, none a multiple of 25; zero overlap (last kg-style line 28 Apr, first bag-style line 2 May).
- **Treatment:** kg = qty × 25 on or after 1 May (239 lines). With it Q3 ties to Finance exactly. Check V11 confirms 0 ambiguous quantities.
- **Residual risk:** the date is proven from the data, **not confirmed by the warehouse**.

DQ-08 · Product-name formatting

- **Found by:** reading the `products` table: `MALTODEXTRIN` has a trailing space and capitals; `sorbitol 70%` is lower-case.
- **Effect:** none on any number, because every join uses `product_id` .
- **Treatment:** not changed in SQL. Product names shown in the report should be tidied for display.

DQ-09 · Lines that look alike but are not exact copies

- **Found by:** supporting queries (appendix Q1, Q2). **714 lines** share an order and product with another line but differ in something else, such as quantity, which is normal. **26**

distributor lines in Q3 share order, product, quantity and discount with another line but differ in price or currency.

- **Treatment:** kept, per Finance (*anything that isn't an exact copy stays*).
- **Open:** distributor orders rose from 47 to 175 between Q1 and Q3, so IT is asked to confirm the 26 are genuine.

DQ-10 · Lines sold below cost (plausibility)

- **Found by:** appendix Q5. **142 lines** have a negative margin, **all in Q3** and none before July:

Group	Lines	Net revenue ₦M
Distributor	66	122.5
Deep-discount	49	15.5
All other	27	8.8
Total	**142**	**146.8**

- **Reading:** consistent with the exchange rate raising costs while prices and discounts moved the other way, so it looks like a business effect and not a data error. It is logged because a negative margin can also come from a mis-keyed price.
- **Open:** Finance to confirm the below-cost sales were intended.

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3. Observations (not defects)

- **Supplier cost steps:** the cost table has 22 rows (20 products plus two changes). P03 and P04 each rose **9.0% on 1 June 2026**. The cost in force on the order date is used.
- **Distributor activity:** all six distributors ordered in January–April and July–September; **five of six** ordered in May and June.
- **Order volume:** 194 orders in January, rising steadily to 226 in September, with no missing month.

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4. Checked and found clean

Check	Result	Where

Orphan keys (lines without order or product; orders without customer)	0, 0, 0	`01` check 1e
Source files agree	The xlsx and CSVs hold identical contents	verified at the start
Key columns empty	None, in all six tables	my runs of `01` to `05`
Duplicate keys in reference tables	None: `customers` 64, `products` 20, `supplier_costs` 22, `fx_rates` 9	my runs of `01` to `05`
Currency values	Only NGN, USD and blank	`01` check 1c
Quantities and prices	Quantity 1 to 3,000; no zero or negative quantity or price	appendix Q3
Discounts	0, 2, 3, 4, 5, then 10 to 13	appendix Q3
Cost and FX coverage	Every line has a supplier cost and an FX rate	V07
One row per line	`line_id` unique after cleaning (4,712)	V06
Reconciliation	Q1 and Q3 tie to Finance exactly	V01 to V04

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5. Not tested

- Customer attributes (segment labels, states) and product categories are taken as given.
- The same customer registered under two IDs was not checked.
- Supplier costs and FX rates were not compared with outside sources.
- **Q2 has no Finance control total**, so April to June are not independently reconciled.
- There are no returns, credit notes or VAT fields, and no invoice totals to compare with.
- **Profiling cannot see meaning:** the bag-unit change was missed until reconciliation. A future profile should show quantity by month for each unit type.

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6. Open items

#	Item	Owner
1	Confirm the 26 near-duplicate distributor lines in Q3 are genuine (DQ-09)	IT
2	Confirm bag products have been recorded in bags since 1 May (DQ-07)	Warehouse
3	Confirm the 142 below-cost lines were intended (DQ-10)	Finance / Commercial
4	Reload `order_lines` once so the raw table holds 4,806 rows (DQ-01)	Me / Data owner
5	Tidy product names for display (DQ-08)	Me

Run record. My DBeaver run (SQLite), 17 October 2026: files 02 to 05 ran without error.

Appendix: supporting queries

Run these after files 02 to 04. They are ad-hoc checks used for this log and are not part of the five SQL files.

```
-- Q1  Lines sharing an order and product with another line (DQ-09). Expect 714
SELECT COUNT(*) AS lines_sharing_order_and_product FROM clean_order_lines l
WHERE (SELECT COUNT(*) FROM clean_order_lines x
        WHERE x.order_id = l.order_id AND x.product_id = l.product_id) > 1;

-- Q2  Distributor Q3 lines sharing order, product, qty and discount with another
SELECT COUNT(*) AS distributor_q3_lines FROM margin_model m
WHERE grp = 'Distributor' AND period = 'Q3'
  AND (SELECT COUNT(*) FROM margin_model x
       WHERE x.order_id = m.order_id AND x.product_id = m.product_id
         AND x.qty = m.qty AND x.discount_pct = m.discount_pct) > 1;

-- Q3  Value ranges. Expect qty 1 to 3000; no zero or negative qty or price;
--     NGN 1012.39 to 28952.31; USD 0.64 to 17.67; discounts 0,2,3,4,5,10,11,12,1
SELECT MIN(qty) AS min_qty, MAX(qty) AS max_qty, MIN(unit_price) AS min_price,
       SUM(qty <= 0) AS qty_zero_or_negative, SUM(unit_price <= 0) AS price_zero_
FROM clean_order_lines;
SELECT currency, MIN(unit_price) AS min_price, MAX(unit_price) AS max_price
```

```
FROM clean_order_lines GROUP BY currency;
SELECT DISTINCT discount_pct FROM clean_order_lines ORDER BY discount_pct;

-- Q4  Date range and orders per month. Expect 2026-01-01 to 2026-09-28; 194, 198
SELECT MIN(order_dt) AS first_day, MAX(order_dt) AS last_day FROM clean_orders;
SELECT month, COUNT(*) AS orders FROM clean_orders GROUP BY month ORDER BY month;

-- Q5  Lines sold below cost (DQ-10). Expect 142 lines, 146.8M
--     (All other Q3 27 / 8.8, Deep-discount Q3 49 / 15.5, Distributor Q3 66 / 12
SELECT grp, period, COUNT(*) AS lines, ROUND(SUM(rev) / 1e6, 1) AS net_rev_M
FROM margin_model WHERE gm < 0 GROUP BY grp, period;
SELECT COUNT(*) AS lines, ROUND(SUM(rev) / 1e6, 1) AS net_rev_M FROM margin_model

-- Q6  Sync-bug duplicates by quarter (DQ-02). Expect Q1 28 / 78.2, Q2 29 / 89.3,
SELECT CASE WHEN o.order_dt BETWEEN '2026-01-01' AND '2026-03-31' THEN 'Q1'
            WHEN o.order_dt BETWEEN '2026-07-01' AND '2026-09-30' THEN 'Q3' ELSE
       COUNT(*) AS duplicate_lines,
       ROUND(SUM(
         CASE WHEN p.unit = 'bag_25kg' AND o.order_dt >= '2026-05-01' THEN d.qty
         * CASE WHEN COALESCE(NULLIF(TRIM(d.currency),''),'NGN') = 'USD'
                THEN d.unit_price * fx.ngn_per_usd ELSE d.unit_price END
         * (1 - d.discount_pct / 100.0)) / 1e6, 1) AS net_rev_M
FROM (
  SELECT *, ROW_NUMBER() OVER (
           PARTITION BY order_id, product_id, qty, unit_price, discount_pct,
                        COALESCE(NULLIF(TRIM(currency),''),'NGN')
           ORDER BY line_id) AS rn
  FROM (SELECT DISTINCT * FROM order_lines)) d
JOIN clean_orders o ON o.order_id = d.order_id
JOIN products p     ON p.product_id = d.product_id
JOIN fx_rates fx    ON fx.month_start = o.month
WHERE d.rn > 1
GROUP BY 1 ORDER BY 1;

-- Q7  Distributors ordering each month. Expect 6,6,6,6,5,5,6,6,6
SELECT month, COUNT(DISTINCT customer_id) AS distributors_ordering
FROM margin_model WHERE grp = 'Distributor' GROUP BY month ORDER BY month;
```